Shiva Shankar S., Ph.D.

bio.shankar@gmail.com | Baltimore, MD, USA

LinkedIn | ORCID 0000-0002-7289-0896 | Google Scholar

Research Summary

Protein designer with 15 years in recombinant protein production, protein engineering and biophysical characterization, who tests his own designs at the bench. Redesigned insoluble mouse Metrnl by ProteinMPNN inverse folding under the constraint that the predicted fold and all five native disulfide bonds be preserved, ranked the candidates computationally, then cloned, expressed and purified them and confirmed activity in a cell-based assay. Designed a *de novo* solubility tag yielding a glycation-resistant insulin proteoform, covered by an issued patent, and published the first structural models of the Meteorin family as first and co-corresponding author. Works in AlphaFold2, Boltz-2, RoseTTAFold, ProteinMPNN, RFdiffusion, BindCraft and PyMOL, with Python, R and Linux. Six peer-reviewed publications.

Education

Ph.D. in Biological Sciences Jul 2017 – Dec 2025
CSIR-National Chemical Laboratory (AcSIR), Pune, India
Thesis: Structural and functional characterization of Meteorin and Meteorin-like proteins
Advisors: Dr. Mahesh J. Kulkarni and Dr. Sureshkumar Ramasamy

M.Sc. in Biotechnology (2009) and **B.Sc. in Biotechnology** (2007)
Bharathiar University, Coimbatore, India

Technical Expertise

Computational Protein Design
– **Inverse folding** and sequence design (**ProteinMPNN**); **RFdiffusion**, **BindCraft**
– Structure prediction (**AlphaFold2**/ColabFold, **Boltz-2**, RoseTTAFold); homology modelling
– Docking, binding-site and **interface analysis**; molecular dynamics (GROMACS, Desmond)

Experimental Validation of Designs
– Expression in *E. coli*, insect and mammalian systems; inclusion-body refolding
– Cloning, construct design, **site-directed mutagenesis**; affinity, IEX and SEC purification
– **Circular dichroism**, SEC-MALS, DLS, **ITC** for fold, assembly, stability and binding

Therapeutic Protein Engineering
– Engineered insulin proteoforms; designed solubility tags; issued patent as first inventor
– Intact-protein and top-down MS; PTM and proteoform characterization

Mass Spectrometry & Structural Methods
– Bottom-up (DDA, DIA/SWATH) and targeted PRM proteomics on Orbitrap Q-Exactive
– Cross-linking MS for structural constraints; crystallization and X-ray diffraction data

Programming & Computing Environment
– **Python**, R for data analysis; **Linux** and **bash** scripting; PyMOL, ChimeraX
– Workflow development and group adoption; curriculum design and scientific training

Fellowships & Honors

- **Chellaram Foundation Diabetes Research Award, 3rd Prize (Basic Science)**, 7th International Diabetes Summit, Pune (2023).
- **CSIR Senior Research Fellowship**, Council of Scientific & Industrial Research (2017). Competitive national fellowship for doctoral research.

Research Experience

Consultant (part-time) Jan 2026 – Present
Cellcraft AI LLC, United States

- Provide protein design, recombinant expression and purification, and structural and functional characterization for protein engineering projects.

Senior Research Fellow / Project Associate Jul 2017 – Dec 2025
CSIR-National Chemical Laboratory, Pune, India
Advisors: Dr. Mahesh J. Kulkarni, Dr. Sureshkumar Ramasamy

Design, Then Experimental Validation

- Redesigned insoluble mouse Metrnl by ProteinMPNN inverse folding while preserving the predicted fold and all five native disulfide bonds, ranked the candidates computationally, then cloned, expressed and purified them; the selected variant expresses soluble in *E. coli* and retains biological activity (manuscript in preparation).
- Measured fold, assembly state and thermal stability on the purified designs by circular dichroism, SEC-MALS, dynamic light scattering and isothermal titration calorimetry, and mapped interfaces by cross-linking mass spectrometry read against the models.
- Established structure-prediction and inverse-folding workflows for the group and trained other members in their use, evaluating and adopting new tools as they appeared.

Therapeutic Protein Engineering

- Designed a *de novo* solubility tag yielding a glycation-resistant insulin proteoform, carrying it from sequence through expression and purification to characterization; first-named inventor on issued patent WO 2024/194897 A1.
- Contributed to the inclusion-body refolding and purification workflow for recombinant proinsulin (*Protein Expression and Purification*, 2025), and characterized modified insulin proteoforms by intact-protein and top-down mass spectrometry.
- Produced the first structural models of the Meteorin protein family with AlphaFold2/ColabFold and RoseTTAFold, resolving domain architecture and putative receptor-binding regions and mapping high-risk missense variants onto them.

Project Assistant II Apr 2013 – Jul 2017
CSIR-National Chemical Laboratory, Pune, India
Advisor: Dr. Hirekodathakallu V. Thulasiram

Recombinant Production and Crystallization

- Cloned, mutagenized and purified class I sesquiterpene synthases from *Santalum album* through multi-step chromatography, then crystallized them by vapour diffusion with robotic screening and collected and screened the resulting X-ray diffraction data.
- Optimized expression, purification and crystallization conditions across several members of the synthase family, working from construct design through to diffraction-quality crystals (*Scientific Reports*, 2015).

Earlier Research Positions Jul 2009 – Apr 2013
IISER-Pune and National Institute of Immunology, India

- *Junior Research Fellow, IISER-Pune (2010–2013):* designed and synthesized short hybrid E-vinylogous lipopeptides whose engineered self-assembly gives broad-spectrum antimicrobial activity (*J. Med. Chem.*, 2013). *Project Trainee, NII New Delhi (2009–2010):* cloned, expressed and purified three *M. tuberculosis* histidine biosynthesis enzymes for crystallographic study.

Publications & Patent

6 peer-reviewed articles | 1 issued patent | 2 as first and co-corresponding author (∗) [Google Scholar]

Patent

1. **Shankar, S.S.***, Kulkarni, M.J., Banarjee, R., Jathar, S., Rajesh, S. "A glycation-resistant insulin polypeptide and a method of preparing the same." *WO 2024/194897 A1*, issued Sep 26, 2024.

Peer-Reviewed Articles

6. Rajesh, S., Jathar, S., Banarjee, R., Sharma, M., Palkar, S., **Shankar, S.S.**, *et al.* "A simple freeze-thaw based method for efficient purification of recombinant human proinsulin from inclusion bodies." *Protein Expression and Purification* **227**, 106645 (2025). doi:10.1016/j.pep.2024.106645

5. **Shankar, S.S.***, Banarjee, R., Jathar, S.M., *et al.* "*De novo* structure prediction of meteorin and meteorin-like protein for identification of domains, functional receptor binding regions, and their high-risk missense variants." *Journal of Biomolecular Structure and Dynamics* **42**(9), 4522–4536 (2024). doi:10.1080/07391102.2023.2220804

4. Pandya, V.K., **Shankar, S.S.**, Sonwane, B.P., *et al.* "Mechanistic insights on anserine hydrolyzing activities of human carnosinases." *Biochimica et Biophysica Acta (BBA) - General Subjects* **1867**(3), 130290 (2023). doi:10.1016/j.bbagen.2022.130290

3. Joshi, R.S., Jagdale, S.S., Bansode, S.B., **Shankar, S.S.**, *et al.* "Discovery of potential multi-target-directed ligands by targeting host-specific SARS-CoV-2 structurally conserved main protease." *Journal of Biomolecular Structure and Dynamics* **39**(9), 3099–3114 (2021). doi:10.1080/07391102.2020.1760137

2. Srivastava, P.L., Daramwar, P.P., Krithika, R., Pandreka, A., **Shankar, S.S.**, & Thulasiram, H.V. "Functional characterization of novel sesquiterpene synthases from Indian sandalwood, Santalum album." *Scientific Reports* **5**, 10095 (2015). doi:10.1038/srep10095

1. **Shankar, S.S.***, Benke, S.N., Nagendra, N., *et al.* "Self-assembly to function: design, synthesis, and broad-spectrum antimicrobial properties of short hybrid E-vinylogous lipopeptides." *Journal of Medicinal Chemistry* **56**(21), 8468–8474 (2013). doi:10.1021/jm400884w

Manuscripts in Preparation

– **Shankar, S.S.**, *et al.* "Inverse folding design enables soluble expression of mouse Metrnl."

– Jathar, S., *et al.*, **Shankar, S.S.** "Glycation-modified insulin glargine by top-down MS."

– **Shankar, S.S.**, *et al.* "Structure-guided discovery of RAGE V-domain antagonists."

Selected Presentations

- *54th American Aging Association (AGE) Meeting*, Utah, USA (2026); *73rd ASMS Conference on Mass Spectrometry*, Baltimore, USA (2025); *24th IUCr Congress of Crystallography*, India (2017).

Mentorship & Professional Service

- **Instructor, CSIR Integrated Skill Initiative** (2021, 2022): wrote and taught the curriculum for two national proteomics programmes covering PRM assay design, Skyline, MaxQuant and FragPipe.
- **Mentorship and service:** supervised 8 project students and trained 6 Ph.D. students; peer reviewer for Analytical Biochemistry, Protein Expression and Purification, and ACS Omega.